

Jake Hui

Brooklyn, NY | 347.222.8815 | jakehuiphone@gmail.com | Portfolio

EDUCATION

New York University

Bachelor of Science in Mechanical Engineering

Jan 2026 – Present

New York, NY

RESEARCH EXPERIENCE

Undergraduate Student Researcher

NYU Prema Lab

Jun 2026 – Present

New York, NY

- Lead CAD design and fabrication of a modular robotic fish to investigate bio-inspired underwater locomotion.
- Develop Python control routines for Dynamixel servos and visualize tail position and motor velocity to support motion testing.

Undergraduate Student Researcher

Biomechatronics & Intelligent Robotics Lab

Jun 2026 – Sep 2026

New York, NY

- Contributed to the design and refinement of a surgical robot for endovascular procedures.
- Created 3D CAD models and mechanical prototypes focused on precise catheter and guidewire manipulation.

WORK EXPERIENCE

Robotics Research Analyst

eFusion Capital

Sep 2025 – Dec 2025

Hong Kong

- Researched public, private, and pre-IPO robotics companies, assessing product capabilities and real-world applications.
- Evaluated technical feasibility and engineering claims to identify limitations and support investment discussions.

Robotic Design Intern

Roborn Technology Ltd

Jul 2024 – Aug 2024

Hong Kong

- Contributed to a functional miniature lunar rover kit for public STEM education in Hong Kong.
- Created 3D models and technical drawings to translate lunar vehicle capabilities into an educational rover design.

ESG Intern

Chun Wo Construction and Engineering Company Ltd

Jul 2023 – Aug 2023

Hong Kong

- Collaborated with the projects division to research ESG solutions addressing environmental challenges and site efficiency.
- Investigated miniature incinerators for on-site waste management, exploring approaches to reduce construction waste.

ENGINEERING PROJECTS & LEADERSHIP

Combat Robotics — Team NEET

Jan 2025 – Present

- Co-developed **13+ robot designs** in Fusion 360 and competed in **4 events**, including NHRL (Feb 2026) with Rem and Ram, a shuffler-based beater bar multibot.

FIRST Tech Challenge — Team 359 WEBB.EXE

2021 – Present

- Co-led the team to **3 Southern California Championships** and **10+ awards**; led mechanical design as co-captain and design lead.
- Mentor mechanical design and team management as an alumni advisor; the team qualified for Worlds in 2025–2026.

Underwater Paleontology Robot

2023 – 2025

- Led a **6-person team** developing an underwater fossil-surveying vehicle for Turlock Lake; secured approximately **\$6,000 in research grants**.
- Collaborated with paleontology researchers at the LA Natural History Museum and Western Science Center.

Accessibility Design Volunteer — The Artemis HK

Jul 2023 – Jan 2026

- Designed an accessible water-entry solution for scuba divers with disabilities and their assistants.

Holonomic Wheelchair — Lumiere Research Program

Jul 2023 – Sep 2023

- Developed and tested a holonomic wheelchair drivetrain; authored design specifications published in the Curieux Academic Journal.

TECHNICAL SKILLS

CAD: Fusion 360, Onshape, SolidWorks

Design & Fabrication: 3D printing, technical drawings, CAM, FEA, robot assembly, maintenance, and repair